Pset 1 – Solutions

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Exercise 1 (Markov Chains)

Let the state space be the set of cities $\{1, 2, 3, 4, 5, 6, 7\}$. From the figure, the (undirected) neighbor sets are

$$N(1) = \{2, 3, 6\}, \quad N(2) = \{1, 4, 7\}, \quad N(3) = \{1, 5\},$$
$$N(4) = \{2, 5, 6, 7\}, \quad N(5) = \{3, 4, 6\}, \quad N(6) = \{1, 4, 5\}, \quad N(7) = \{2, 4\}.$$

Hence the degrees are

$$\deg(1) = 3,\ \deg(2) = 3,\ \deg(3) = 2,\ \deg(4) = 4,\ \deg(5) = 3,\ \deg(6) = 3,\ \deg(7) = 2,$$

and $\sum_{j=1}^{7} \deg(j) = 20$.

(a) **Prove that (X_j) is a Markov chain.**

By construction: given $X_j = i$, the next city X_{j+1} is chosen uniformly from the neighbors $N(i)$, and does not depend on earlier history.

Formally, for any $j \geq 0$ and any states i, k,

$$\mathbb{P}(X_{j+1} = k \mid X_j = i, X_{j-1}, \ldots, X_0) = \mathbb{P}(X_{j+1} = k \mid X_j = i) = \begin{cases} \frac{1}{\deg(i)}, & k \in N(i), \\ 0, & k \notin N(i). \end{cases}$$

This is the Markov property, so (X_j) is a Markov chain.

(b) **What is the transition matrix P?**

Index states in the order $(1, 2, 3, 4, 5, 6, 7)$. Then $P_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i)$ equals $1/\deg(i)$ if $j \in N(i)$, else 0. Thus

$$P = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}.$$

(c) **Prove that the stationary distribution is $\pi(j) = \deg(j)/20$.**

Define

$$\pi(j) = \frac{\deg(j)}{\sum_{\ell=1}^{7} \deg(\ell)} = \frac{\deg(j)}{20}.$$

We verify $\pi P = \pi$. For any fixed j,

$$(\pi P)(j) = \sum_{i=1}^{7} \pi(i) P_{ij} = \sum_{i \in N(j)} \pi(i) \frac{1}{\deg(i)} = \sum_{i \in N(j)} \frac{\deg(i)}{20} \cdot \frac{1}{\deg(i)} = \sum_{i \in N(j)} \frac{1}{20} = \frac{\deg(j)}{20} = \pi(j).$$

So π is stationary.

(d) **Prove that the chain is reversible: $\pi(i)P_{ij} = \pi(j)P_{ji}$.**

If i and j are not neighbors, then $P_{ij} = P_{ji} = 0$, so the equality holds.

If $i \sim j$ (there is an edge), then

$$\pi(i)P_{ij} = \frac{\deg(i)}{20} \cdot \frac{1}{\deg(i)} = \frac{1}{20} = \frac{\deg(j)}{20} \cdot \frac{1}{\deg(j)} = \pi(j)P_{ji}.$$

Hence the chain is reversible.

(e) **Lazy walk: stay with probability $\alpha \in (0,1)$. Does π change?**

The new transition matrix is

$$P^{(\alpha)} = \alpha I + (1-\alpha)P.$$

If π is stationary for P, i.e. $\pi P = \pi$, then

$$\pi P^{(\alpha)} = \pi\big(\alpha I + (1-\alpha)P\big) = \alpha\pi + (1-\alpha)\pi P = \alpha\pi + (1-\alpha)\pi = \pi.$$

So the stationary distribution does *not* change: it remains $\pi(j) = \deg(j)/20$.

Exercise 2 (Linear Algebra)

From the figure, the undirected edges are

$$\{(1,2),(1,3),(2,3),(2,4),(3,5),(4,5),(4,6),(5,6),(4,7),(6,7)\}.$$

Degrees:

$$\deg(1) = 2,\ \deg(2) = 3,\ \deg(3) = 3,\ \deg(4) = 4,\ \deg(5) = 3,\ \deg(6) = 3,\ \deg(7) = 2.$$

(a) **Adjacency matrix A and Laplacian $L = D - A$.**

With vertex order $(1,2,3,4,5,6,7)$,

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \qquad D = \operatorname{diag}(2,3,3,4,3,3,2).$$

Thus

$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 & 0 & 0 \\ -1 & -1 & 3 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 4 & -1 & -1 & -1 \\ 0 & 0 & -1 & -1 & 3 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 3 & -1 \\ 0 & 0 & 0 & -1 & 0 & -1 & 2 \end{pmatrix}.$$

(b) **Compute the eigenvalues of L. Remark that 1 is an eigenvector.**

Numerically (e.g. using Python for the symmetric matrix L), the eigenvalues are approximately

$$\lambda(L) \approx \{0,\ 0.814349,\ 2.328009,\ 3.313908,\ 3.598089,\ 4.457530,\ 5.488115\}.$$

Also,

$$L\mathbf{1} = 0 \quad \text{where } \mathbf{1} = (1,1,1,1,1,1,1)^\top,$$

because each row of L sums to 0. Hence $\mathbf{1}$ is an eigenvector with eigenvalue 0.

(c) **For a general undirected graph, prove that for any $x \in \mathbb{R}^n$, $x^\top L x = \sum_{(i,j)\in E}(x_i - x_j)^2$.**

For an undirected graph,

$$x^\top L x = x^\top (D - A)x = \sum_i \deg(i) x_i^2 - \sum_{i,j} A_{ij} x_i x_j.$$

Rewrite the first term as $\sum_{i,j} A_{ij} x_i^2$ (since $\deg(i) = \sum_j A_{ij}$):

$$x^\top L x = \sum_{i,j} A_{ij} x_i^2 - \sum_{i,j} A_{ij} x_i x_j = \frac{1}{2}\sum_{i,j} A_{ij}\big(x_i^2 + x_j^2 - 2x_i x_j\big) = \frac{1}{2}\sum_{i,j} A_{ij}(x_i - x_j)^2.$$

Each undirected edge $(i,j) \in E$ appears twice in $\sum_{i,j}$, so

$$x^\top L x = \sum_{(i,j)\in E} (x_i - x_j)^2.$$

(d) **Use this identity to prove that L is positive semidefinite and that the smallest eigenvalue is always $\lambda_1 = 0$.**

From (c),

$$x^\top L x = \sum_{(i,j)\in E} (x_i - x_j)^2 \geq 0 \quad \text{for all } x,$$

so $L \succeq 0$ (positive semidefinite), hence all eigenvalues are ≥ 0. But $L\mathbf{1} = 0$, so 0 is an eigenvalue. Therefore the smallest eigenvalue is

$$\lambda_1 = 0.$$

(e) **Let λ_2 be the second-smallest eigenvalue. Prove that if G is disconnected, then $\lambda_2 = 0$.**

Assume G has at least two connected components, and let C be one component. Let $\mathbf{1}_C \in \mathbb{R}^n$ be the indicator vector:

$$(\mathbf{1}_C)_i = \begin{cases} 1, & i \in C, \\ 0, & i \notin C. \end{cases}$$

Because there are no edges between C and C^c, every edge $(i,j) \in E$ has both endpoints in the same component, hence $(\mathbf{1}_C)_i - (\mathbf{1}_C)_j = 0$. By (c),

$$\mathbf{1}_C^\top L \mathbf{1}_C = \sum_{(i,j)\in E} \big((\mathbf{1}_C)_i - (\mathbf{1}_C)_j\big)^2 = 0,$$

so $\mathbf{1}_C \in \ker(L)$. Also $\mathbf{1}_C$ is not a multiple of $\mathbf{1}$, so $\dim\ker(L) \geq 2$, meaning eigenvalue 0 has multiplicity at least 2. Therefore the second-smallest eigenvalue is

$$\lambda_2 = 0.$$

Exercise 3 (Statistical Inference)

Let $X_1, \ldots, X_n$ be i.i.d. with density

$$f(x) = \frac{1}{2\sigma} e^{-|x|/\sigma}, \qquad \sigma > 0.$$

(a) **Write out the log-likelihood function $\ell(X_1, \ldots, X_n)$.**

The likelihood is

$$L(\sigma) = \prod_{i=1}^{n} \frac{1}{2\sigma} \exp\left(-\frac{|X_i|}{\sigma}\right).$$

Thus the log-likelihood is

$$\ell(\sigma) = \log L(\sigma) = \sum_{i=1}^{n} \left[-\log(2\sigma) - \frac{|X_i|}{\sigma}\right] = -n\log(2\sigma) - \frac{1}{\sigma}\sum_{i=1}^{n} |X_i|.$$

(b) **What is the MLE of σ?**

Differentiate for $\sigma > 0$:

$$\ell'(\sigma) = -\frac{n}{\sigma} + \frac{1}{\sigma^2}\sum_{i=1}^{n} |X_i|.$$

Set $\ell'(\sigma) = 0$:

$$-\frac{n}{\sigma} + \frac{1}{\sigma^2}\sum_{i=1}^{n} |X_i| = 0 \quad \Longleftrightarrow \quad -n\sigma + \sum_{i=1}^{n} |X_i| = 0 \quad \Longleftrightarrow \quad \hat{\sigma} = \frac{1}{n}\sum_{i=1}^{n} |X_i|.$$

Second derivative:

$$\ell''(\sigma) = \frac{n}{\sigma^2} - \frac{2}{\sigma^3}\sum_{i=1}^{n} |X_i|,$$

and at $\sigma = \hat{\sigma}$, $\ell''(\hat{\sigma}) = \frac{n}{\hat{\sigma}^2} - \frac{2n\hat{\sigma}}{\hat{\sigma}^3} = -\frac{n}{\hat{\sigma}^2} < 0$, so $\hat{\sigma}$ is indeed a maximizer.

(c) **Is the MLE consistent? Prove your answer.**

Compute $\mathbb{E}|X_1|$. By symmetry,

$$\mathbb{E}|X_1| = 2\int_0^\infty x \cdot \frac{1}{2\sigma} e^{-x/\sigma}\, dx = \frac{1}{\sigma}\int_0^\infty x e^{-x/\sigma}\, dx.$$

Let $y = x/\sigma$ so $x = \sigma y$, $dx = \sigma dy$:

$$\mathbb{E}|X_1| = \frac{1}{\sigma}\int_0^\infty (\sigma y) e^{-y}\, \sigma dy = \sigma \int_0^\infty y e^{-y}\, dy = \sigma \cdot 1 = \sigma,$$

since $\int_0^\infty y e^{-y}\, dy = 1$.

Now

$$\hat{\sigma} = \frac{1}{n}\sum_{i=1}^{n} |X_i|.$$

By the strong law of large numbers,

$$\hat{\sigma} \xrightarrow{a.s.} \mathbb{E}|X_1| = \sigma,$$

so $\hat{\sigma}$ is consistent.

Exercise 4 (Probability Theory)

Assume the standard (undirected) graphon sampling: draw $U_1, \dots, U_n \overset{\text{iid}}{\sim} \text{Unif}(0,1)$, and (conditionally on the U's) the edges $\{a_{ij} : 1 \le i < j \le n\}$ are independent with

$$a_{ij} \mid (U_i, U_j) \sim \text{Bern}(W(U_i, U_j)), \qquad a_{ji} = a_{ij}, \qquad a_{ii} = 0.$$

Define degrees $d_i = \sum_{j \neq i} a_{ij}$.

Introduce the graphon degree function

$$g(u) := \int_0^1 W(u,v)\, dv,$$

and the global edge density

$$\mu := \int_0^1 \int_0^1 W(u,v)\, du\, dv = \mathbb{E}[W(U_1, U_2)].$$

(a) **Compute $\mathbb{E}[d_i \mid U_i]$ and $\mathbb{E}[d_i]$.**

For $j \neq i$,

$$\mathbb{E}[a_{ij} \mid U_i] = \mathbb{E}[\mathbb{E}[a_{ij} \mid U_i, U_j] \mid U_i] = \mathbb{E}[W(U_i, U_j) \mid U_i] = \int_0^1 W(U_i, v)\, dv = g(U_i).$$

Therefore

$$\mathbb{E}[d_i \mid U_i] = (n-1)\, g(U_i).$$

Taking expectation again,

$$\mathbb{E}[d_i] = (n-1)\, \mathbb{E}[g(U_i)] = (n-1)\mu.$$

(b) **Compute $\text{Var}(d_i \mid U_i)$ and $\text{Var}(d_i)$.**

Given U_i, the random variables $\{a_{ij} : j \neq i\}$ are independent and each satisfies

$$\mathbb{P}(a_{ij} = 1 \mid U_i) = g(U_i),$$

so $a_{ij} \mid U_i \sim \text{Bern}(g(U_i))$. Hence

$$\text{Var}(a_{ij} \mid U_i) = g(U_i)\big(1 - g(U_i)\big),$$

and summing independent terms,

$$\text{Var}(d_i \mid U_i) = (n-1)\, g(U_i)\big(1 - g(U_i)\big).$$

For the unconditional variance, use the law of total variance:

$$\text{Var}(d_i) = \mathbb{E}[\text{Var}(d_i \mid U_i)] + \text{Var}(\mathbb{E}[d_i \mid U_i]).$$

Thus

$$\text{Var}(d_i) = (n-1)\, \mathbb{E}\big[g(U)(1-g(U))\big] + (n-1)^2\, \text{Var}(g(U)).$$

(c) **Compute** $\mathrm{Cov}(d_i, d_j)$ **for** $i \neq j$**.**

Write

$$d_i = a_{ij} + \sum_{k \neq i,j} a_{ik}, \qquad d_j = a_{ij} + \sum_{k \neq i,j} a_{jk}.$$

Let

$$\gamma := \mathbb{E}\big[g(U)^2\big] = \mathbb{E}[W(U_1, U_2)\, W(U_1, U_3)] \quad (\text{with } U_1, U_2, U_3 \text{ } iid).$$

Then

$$\mathbb{E}[a_{ij}] = \mu, \quad \mathrm{Var}(a_{ij}) = \mu(1-\mu),$$

and for any $k \neq i, j$,

$$\mathrm{Cov}(a_{ij}, a_{ik}) = \gamma - \mu^2, \quad \mathrm{Cov}(a_{ij}, a_{jk}) = \gamma - \mu^2, \quad \mathrm{Cov}(a_{ik}, a_{jk}) = \gamma - \mu^2.$$

There are $(n-2)$ choices of k, so

$$\mathrm{Cov}(d_i, d_j) = \mu(1-\mu) \; + \; 3(n-2)\,(\gamma - \mu^2).$$

(d) **Limits of** d_1/n **and** d_2/n**. Do they converge to the same quantity? CLT?**

Condition on U_1. Since $d_1 \mid U_1 \sim \mathrm{Bin}(n-1, g(U_1))$,

$$\frac{d_1}{n-1} \xrightarrow{a.s.} g(U_1), \qquad \text{hence} \qquad \frac{d_1}{n} \xrightarrow{a.s.} g(U_1).$$

Similarly,

$$\frac{d_2}{n} \xrightarrow{a.s.} g(U_2).$$

In general, they do *not* converge to the same quantity: $g(U_1)$ and $g(U_2)$ are i.i.d. and are almost surely equal only if g is (a.e.) constant.

CLT (conditional). For fixed u, if $d_1 \mid U_1 = u \sim \mathrm{Bin}(n-1, g(u))$, then

$$\frac{d_1 - (n-1)g(u)}{\sqrt{(n-1)g(u)(1-g(u))}} \xrightarrow{d} \mathcal{N}(0,1).$$

Equivalently, conditioning on U_1,

$$\sqrt{n}\left(\frac{d_1}{n} - g(U_1)\right) \;\Big|\; U_1 \;\xrightarrow{d}\; \mathcal{N}(0,\; g(U_1)(1-g(U_1)))\,.$$

Unconditionally this yields a mixture of normals (variance depends on the random U_1).

(e) **Limit of** $\dfrac{\sum_{i \leq n} d_i}{n^2}$**.**

In an undirected graph,

$$\sum_{i=1}^{n} d_i = 2 \sum_{1 \leq i < j \leq n} a_{ij}.$$

Thus

$$\frac{1}{n^2} \sum_{i=1}^{n} d_i = \frac{2}{n^2} \sum_{i<j} a_{ij}.$$

Decompose $a_{ij} = (a_{ij} - W(U_i, U_j)) + W(U_i, U_j)$. The first part averages to 0 and the second part is a U-statistic that converges almost surely to $\mu = \iint W$. Therefore

$$\frac{1}{n^2} \sum_{i=1}^{n} d_i \xrightarrow{a.s.} \iint_{[0,1]^2} W(u,v)\, du\, dv.$$